NLU course project

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1. Introduction

In this project, I start from a baseline recurrent language model implemented as RNN[1][2] and applied a series of incremental enhancements to reduce validation perplexity (PPL) on the Penn Treebank dataset.

The first part explores replacing the recurrent cell with an LSTM, adding dropout at strategic positions, and switching the optimizer from SGD to AdamW, with hyperparameter tuning focused on learning rate.

The second part builds upon the best LSTM model, applying weight tying, variational dropout, and a non-monotonically triggered averaged SGD regime.

The goal is to achieve perplexity below 250 on the validation set.

2. Implementation details

For the first part, some incremental improvements are made, and for each of them experiments were taken and tracked. First of all the Recurrent Neural Network was substituted with a Long Short Term Memory.
After that, two dropout layers were inserted immediately after the embedding layer and just before the final layer. Dropout layers, during training, randomly zeroes some of the elements of the input tensor with some probability. This step is done to counteract overfitting.

An important task done in addition to optimization techniques is hyperparameter optimization. First of all I executed different experiments to tune batch size, learning rate, hidden size and embedding size. After these tries I choose values that minimizes perplexity and then i keep them for future experiments. The model is trained up to 100 epochs with early stopping based on validation perplexity.

Following these architectural changes, I replaced the original SGD optimizer with AdamW, which decouples weight decay from the gradient update and often yields faster convergence. After changing the optimizer, it was necessary to tune the learning rate again.

All of these improvements make the model perform better, so all of these were maintained.

In the second phase, I took the optimized LSTM as a backbone and added three advanced techniques inspired by the paper by Merity et al. [3].
The model also incorporates weight tying, a technique where the weights of the output softmax layer are shared with the input embedding layer. This method, not only reduces the number of parameters but also encourages the model to learn a more coherent shared representation of words. Weight tying helps regularize the model and improves performance, particularly in tasks with large vocabularies. To apply this method, dimensions of hidden layer and embedding layer must be the same, so I have added a control that activate weight tying only if these two dimensions are equal.
Traditional dropout applies a different dropout mask at each time step, which can destabilize recurrent networks like LSTMs. To address this it can be used variational dropout which applies the same dropout mask across all time steps for a given layer and batch. This consistent masking encourages the network to rely less on specific neurons and generalize better, thereby reducing overfitting.
In addition to regularization techniques like variational dropout and weight tying, the model incorporates an optimization strategy called Non-monotonically Triggered Averaged Stochastic Gradient Descent (NT-ASGD). This method is a variation of traditional Stochastic Gradient Descent (SGD) that averages the model parameters over several steps to improve generalization. NT-ASGD introduces an adaptive mechanism: it monitors validation performance during training and only starts averaging when the validation metric stops improving for a fixed number of evaluations. This delay ensures that averaging begins only when the model has reached a stable region in the parameter space. The averaging is done over consecutive updates after the trigger point and continues until the end of training.

3. Results

The performance of each experiment was evaluated using validation perplexity (PPL) as the main metric.

In the first part (Table 1), starting from a baseline RNN trained with SGD, successive architectural and optimization enhancements were applied. Replacing the vanilla RNN with an LSTM cell already resulted in a significant drop in perplexity, from 154.33 to 137.49. Introducing dropout layers after the embedding and before the output linear layer further improved performance, reducing the PPL to 118.92. However, replacing SGD with AdamW did not yield better results, increasing perplexity slightly to 119.91.

In the second part (Table 2), starting from the best LSTM model, further improvements were made using regularization techniques. Weight tying (WT) alone led to a modest improvement, bringing validation perplexity to 114.56. Adding variational dropout (VD) provided a substantial reduction in perplexity, achieving 104.38. Finally, incorporating non-monotonically triggered averaged SGD (NT-ASGD) slightly enhanced the results further, achieving the best validation perplexity of 104.22. Figures 1 and 2 illustrate the evolution of the loss and perplexity during training for the best configuration (LSTM + WT + VD + NTAvSGD). The figures highlight the point where NTvASGD is activated, after the validation metrics plateaued.

Model configuration	Validation PPL
Baseline RNN (SGD)	154.33
LSTM only	137.49
+ Dropout layers	118.92
+ AdamW	119.91

Table 1: *Validation perplexity (PPL) for each experiment of First Part*

Model configuration	Validation PPL
LSTM WT	114.56
LSTM WT + VD	104.38
LSTM WT + VD + NTAvSGD	104.22

Table 2: *Validation perplexity (PPL) for each experiment of Second Part*

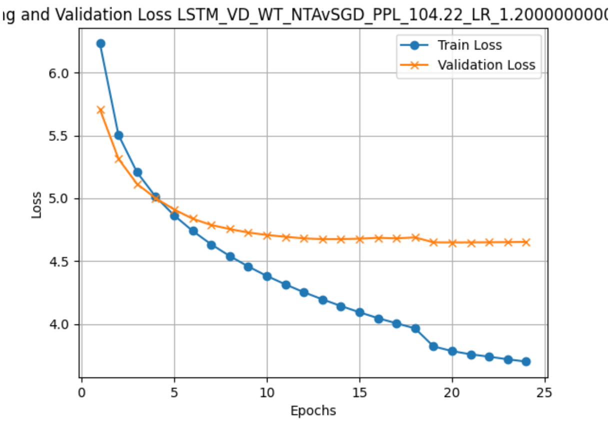

Figure 1: *Loss plot of the best performance of Second Part. It is visible the point where model switch to NTAvSGD*

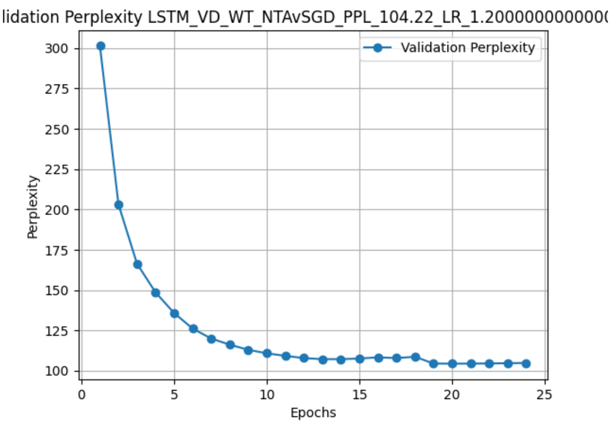

Figure 2: *Perplexity plot of the best performance of Second Part. It is visible the point where model switch to NTAvSGD*

4. References

[1] T. Mikolov, S. Kombrink, L. Burget, J. Černocký, and S. Khudanpur, "Extensions of recurrent neural network language model," in *2011 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2011, pp. 5528–5531.

[2] T. Mikolov, M. Karafiát, L. Burget, J. Cernockỳ, and S. Khudanpur, "Recurrent neural network based language model." in *Interspeech*, vol. 2, no. 3. Makuhari, 2010, pp. 1045–1048.

[3] S. Merity, N. S. Keskar, and R. Socher, "Regularizing and Optimizing LSTM Language Models," *arXiv preprint arXiv:1708.02182*, 2017.